

Golden-ratio growth of Conway's subprime closure

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Abstract

Let $s(m)$ be the Conway subprime function [8], [10], define the binary operation on the natural numbers $x \circ y = s(x + y)$, and denote by C_n , $n \geq 0$, the sequence of subsets of natural numbers defined by $C_0 = \{1\}$, and $C_{n+1} = C_n \cup (C_n \circ C_n)$. We prove the conjecture [2] by Caragiu, Vicol and Zaki that

$$\lim_{n \rightarrow \infty} \frac{|C_{n+1}|}{|C_n|} = \frac{1 + \sqrt{5}}{2}.$$

The underlying mathematical proof in this paper was constructed with some algorithmic assistance from GPT-6 Astra [9]. The main theorem has been formally verified in the Lean proof assistant, taking as analytic input the Siegel–Walfisz theorem and a Vinogradov-type minor-arc bound; the formalization is available at [4].

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1 Introduction

For a positive integer m , the Conway's subprime function [8], [10] is defined as follows

$$s(m) = \begin{cases} 1, & m = 1, \\ m, & m \text{ is prime,} \\ m/\text{lpf}(m), & m \text{ is composite,} \end{cases}$$

where $\text{lpf}(m)$ is the least prime factor of m .

The Conway subprime function is used in the construction of subprime Fibonacci sequences, i.e. integer sequences $\{x_k\}$, $k \geq 0$, satisfying a recursion $x_k = s(x_{k-1} + x_{k-2})$, which have been studied (see [5]) for periodicity, in analogy with the Collatz problem.

In [2], Caragiu, Vicol, and Zaki use the Conway subprime function to construct the following sets. Starting with 1, retain all previously generated numbers and apply the operation $(a, b) \mapsto s(a + b)$ to every pair available at the preceding stage:

$$C_0 = \{1\}, \quad C_{n+1} = C_n \cup (C_n \circ C_n) = C_n \cup \{s(a + b) : a, b \in C_n\}. \quad (1.1)$$

The main result of [2] is that

$$\bigcup_{n \geq 0} C_n = \mathbb{N}, \quad (1.2)$$

The golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ makes an unexpected appearance in their paper, since Caragiu, Vicol, and Zaki conjecture in [2] that

$$\lim_{n \rightarrow \infty} \frac{|C_{n+1}|}{|C_n|} = \varphi. \quad (1.3)$$

Write $M_n = \max C_n$. For $n \geq 1$, let R_n be the first prime absent from C_n , and let Q_n be the prime immediately preceding R_n . These are well-defined: C_n is finite and contains 2. Thus every prime at most Q_n belongs to C_n , while $R_n \notin C_n$ and $Q_n \leq M_n$. We prove the following theorem, which is a more precise version of the aforementioned conjecture 3 in [2]:

Theorem 1.1. *There is a constant $c > 0$ such that*

$$M_n \sim Q_n \sim c\varphi^n, \quad |C_n| \sim c\varphi^{n-1}. \quad (1.4)$$

Consequently,

$$\lim_{n \rightarrow \infty} \frac{|C_{n+1}|}{|C_n|} = \varphi, \quad \lim_{n \rightarrow \infty} \frac{|C_n|}{M_n} = \frac{1}{\varphi}. \quad (1.5)$$

The proof of the theorem uses the cited exhaustion result in [2], an almost-all theorem for Goldbach representations with nearly equal primes, and two standard estimates for exponential sums over primes. The particular interval-restricted representation lemma needed here is proved in Appendix A. The proof was constructed with algorithmic assistance from OpenAI Astra. A Lean formalization of the main theorem, assuming the Siegel–Walfisz theorem and a Vinogradov-type minor-arc bound, is available at [4].

2 Heuristics and Proof overview

1. **Parity gives the Fibonacci upper bound.** If a sum is composite, division by its least prime factor prevents the output from exceeding the preceding maximum. A new maximum must therefore be prime. An odd prime is a sum of an odd and an even input, and the even input cannot exceed the maximum from one generation earlier. This gives $M_{n+1} \leq M_n + M_{n-1}$.
2. **Complete prime prefixes reproduce at the Fibonacci scale.** When every prime up to X is present, almost-all Goldbach representations produce all but $O_A(X/\log^A X)$ integers up to X one generation later. These integers, paired with primes near a larger cutoff, extend the complete prime prefix by an almost Fibonacci step. Summable relative losses then give positive constants c, d with $M_n \sim c\varphi^n$ and $Q_n \sim d\varphi^n$.
3. **A gap between these constants produces extra primes.** If $d < c$, an earlier maximum supplies a generated prime t beyond the predicted complete prefix at a suitable scale. The equation $2p + q = 2u - t$, with p and q in specified complete prime intervals, generates u through the two operations $(t + q)/2$ and $p + (t + q)/2$. An almost-all representation estimate therefore produces almost every prime in an interval beyond the predicted prefix.
4. **Two small exceptional sets eliminate the gap.** Take the first missing prime two generations later. It has many possible decompositions $r = u + (r - u)$ with u in the extra interval. Only a small number of these u are missing, and only a small number of their complementary integers $r - u$ are missing. At least one pair is present, forcing r to be generated and contradicting its definition. Hence $d = c$.

5. **The preceding maximum determines the cardinality.** Since $Q_{n-1} \sim M_{n-1}$, the Goldbach filling estimate makes C_n almost complete below M_{n-1} . Every element above that threshold is prime, and those primes contribute only $o(M_{n-1})$ elements. Thus $|C_n| \sim M_{n-1}$, yielding both the golden-ratio growth and the limiting density $1/\varphi$.

3 Analytic preliminaries

All logarithms are natural. We will denote cardinality of a set X either by $|X|$ or by $\#X$. The notations $f = O_A(g)$ and $f \ll_A g$ mean $|f| \leq C_A g$ for all sufficiently large values of the relevant parameter. We will write $f \asymp g$ when both $f \ll g$ and $g \ll f$ hold, and $f \sim g$ when $f/g \rightarrow 1$.

Let $\mathbb{P}$ denote the set of primes and $\pi(X)$ the number of primes at most X . Intervals used to count missing elements are to be intersected with $\mathbb{N}$. Define

$$H_j(X) = \#(\{1, \dots, \lfloor X \rfloor\} \setminus C_j).$$

Lemma 3.1 (Prime counts). *As $X \rightarrow \infty$,*

$$\# \left(\mathbb{P} \cap \left[X - \frac{X}{\log^2 X}, X \right] \right) \sim \frac{X}{\log^3 X}. \quad (3.1)$$

For fixed $0 < \alpha < \beta$,

$$\#(\mathbb{P} \cap [\alpha X, \beta X]) \sim \frac{(\beta - \alpha)X}{\log X}. \quad (3.2)$$

Also $\pi(X) \sim X/\log X$. The prime immediately preceding a large real X is $X + O(X/\log^2 X)$, and the prime immediately following X is $X + o(X)$.

Proof. The $q = 1$, zero-frequency case of [7, Lemma 5 (Major arc estimates), equation (26)] or [11, Proposition 24], which both follow from the Siegel-Walfisz theorem on primes in arithmetic progressions, yields for every fixed $D > 0$, that

$$\sum_{m \leq X} \Lambda(m) = X + O_D(X/\log^D X).$$

The prime powers with exponent at least two have total weight $O(\sqrt{X} \log^2 X)$: there are at most $\log_2 X$ possible exponents, and for each exponent at most $\sqrt{X}$ bases, each of weight at most $\log X$. Thus

$$\vartheta(X) := \sum_{p \leq X} \log p = X + O_D(X/\log^D X). \quad (3.3)$$

Subtract this formula at X and $X - h$, where $h = X/\log^2 X$, taking $D = 4$. It gives total prime weight $h + O(X/\log^4 X) \sim h$ in $(X - h, X]$. Every such prime has logarithm $\sim \log X$, uniformly, which proves (3.1); changing an endpoint adds at most one prime. The same subtraction at βX and αX proves (3.2). Partial summation gives

$$\pi(X) = \frac{\vartheta(X)}{\log X} + \int_2^X \frac{\vartheta(t)}{t \log^2 t} dt = \frac{X}{\log X} + O(X/\log^2 X).$$

For the integral bound, split at $\sqrt{X}$ and use $\vartheta(t) \ll t$. The preceding-prime claim follows from the nonemptiness of the interval in (3.1). For every fixed $\epsilon > 0$, (3.2) supplies a prime between X and $(1 + \epsilon)X$ for all sufficiently large X . Letting ϵ be arbitrary proves the following-prime claim. $\square$

Lemma 3.2 (Goldbach filling). *Fix $A > 0$. If C_j contains every prime at most X , then*

$$H_{j+1}(X) \ll_A \frac{X}{\log^A X}. \quad (3.4)$$

The implicit constant is independent of j .

Proof. The corollary to Theorem 1 in [3], with exponent $5/8 + 1/8 = 3/4$, or equations (1.5), (1.6) and Theorem 1.2 in [1] yield that all but $O_A(Y/\log^A Y)$ integers $m \in [Y, 2Y]$ have a representation

$$2m = p + q, \quad p, q \in \mathbb{P}, \quad |p - m|, |q - m| \leq m^{3/4}. \quad (3.5)$$

Here their parameter for the even number $2m$ is rescaled by a factor of two, which does not change the order of the exceptional-set bound.

Consider $\sqrt{X} < m \leq X - X^{3/4}$. Cover this range by dyadic intervals $[X/2^{r+1}, X/2^r]$ that meet it. Each lower endpoint Y is at least $\sqrt{X}/2$, their sum is at most X , and therefore $\log Y \geq \frac{1}{3} \log X$ for large X . The total number of exceptions to (3.5) is consequently at most

$$C_A \sum_r \frac{X/2^{r+1}}{\log^A(X/2^{r+1})} \leq 3^A C_A \frac{X}{\log^A X}.$$

For every nonexceptional m in this range, $p, q \leq m + m^{3/4} \leq X$. Hence $p, q \in C_j$, and, as $m \geq 2$, the number $2m$ is composite with least prime factor 2. Thus $s(p + q) = m \in C_{j+1}$. The omitted bottom and top ranges contain $O(\sqrt{X} + X^{3/4} + 1)$ integers. This is $O_A(X/\log^A X)$, proving the lemma. $\square$

Lemma 3.3 (Two primes in specified intervals). *Fix $0 < \alpha < \beta$, $0 < \gamma < \delta$, $\eta > 0$, $B > 0$, and $A > 0$. Put*

$$I_X = [\alpha X, \beta X], \quad J_X = [\gamma X, \delta X], \quad \mathcal{J}_X(N) = \text{meas}\{v \in I_X : N - 2v \in J_X\}.$$

Among the odd integers $1 \leq N \leq BX$ with $\mathcal{J}_X(N) \geq \eta X$, at most $O(X/\log^A X)$ fail to have a representation

$$N = 2p + q, \quad p \in \mathbb{P} \cap I_X, \quad q \in \mathbb{P} \cap J_X. \quad (3.6)$$

The implicit constant may depend on all the displayed fixed parameters.

The proof of this lemma is given in Appendix A. It uses no assertion about every individual odd target; the exceptional set is essential.

4 Fibonacci growth and the two asymptotic constants

Lemma 4.1 (Structure and upper bound). *For $n \geq 1$, the maximum M_n is prime and*

$$C_n \subseteq \{1, \dots, M_{n-1}\} \cup (\mathbb{P} \cap [1, M_n]). \quad (4.1)$$

For $n \geq 1$,

$$M_{n+1} \leq M_n + M_{n-1}, \quad M_n \leq F_{n+2}, \quad (4.2)$$

where $F_0 = 0$, $F_1 = 1$, and $F_{r+2} = F_{r+1} + F_r$.

Proof. If $a, b \leq M_{n-1}$ and $a + b$ is composite, then

$$s(a + b) = \frac{a + b}{\text{lpf}(a + b)} \leq \frac{a + b}{2} \leq M_{n-1}.$$

If $a + b$ is prime, the output is prime. Previously retained elements are also at most M_{n-1} , proving (4.1). Since $M_1 = 2$, a maximum that changes can change only to a prime, and a maximum that does not change remains prime. Moreover $M_2 = 3$, so M_n is odd for $n \geq 2$.

For $n \geq 2$, every even element of C_n is at most M_{n-1} : this follows from (4.1) for composites, and from $2 \leq M_{n-1}$ for the only even prime. If $M_{n+1} > M_n$, its generating sum is an odd prime, so one input is even and at most M_{n-1} , and the other is at most M_n . If the maximum does not increase, the same bound is immediate. The case $n = 1$ is $M_2 = 3 = M_1 + M_0$. Induction from $M_0 = 1$, $M_1 = 2$ now proves the Fibonacci bound. In particular $M_n \ll \varphi^n$, for example by the formula $F_r = (\varphi^r - (-\varphi)^{-r})/\sqrt{5}$. $\square$

Lemma 4.2 (Extension of a complete prime prefix). *Fix $B \geq 1$. Suppose $Y \leq X \leq BY$, C_n contains every prime at most X , and C_{n-1} contains every prime at most Y . For sufficiently large Y , depending only on B , the set C_{n+1} contains every prime at most*

$$X + Y - \frac{X}{\log^2 X}. \quad (4.3)$$

Proof. Set $h = X/\log^2 X$. For large Y , $h < Y$. Lemma 3.2 gives $H_n(Y) \ll Y/\log^4 Y$. Let r be a prime with $X < r \leq X + Y - h$. For each prime $p \in [X - h, X]$, the integer $a = r - p$ is positive and satisfies

$$a \leq (X + Y - h) - (X - h) = Y.$$

Distinct choices of p give distinct a . By Lemma 3.1, the number of candidate primes is asymptotic to $X/\log^3 X$, which exceeds $H_n(Y)$ for sufficiently large Y : indeed $X \asymp_B Y$ and $\log X \sim \log Y$. Some candidate therefore has $a \in C_n$. Also $p \in C_n$, and $s(p + a) = r$ because r is prime. The primes at most X are retained, completing the proof. $\square$

Proposition 4.3. *We have $Q_n \asymp M_n \asymp \varphi^n$.*

Proof. Choose a fixed real L large enough for Lemma 4.2 with $B = 2$, and such that $1/\log^2 L \leq 1/4$. By (1.2) and nesting there is an N with $\{1, \dots, \lceil L \rceil\} \subseteq C_N$. Define

$$L_0 = L_1 = L, \quad L_{k+1} = L_k + L_{k-1} - \epsilon_k L_k, \quad \epsilon_k = \frac{1}{\log^2 L_k} \quad (k \geq 1). \quad (4.4)$$

Inductively $L_{k-1} \leq L_k \leq 2L_{k-1}$ and $L_k \geq L$. Indeed, under these inequalities, $\epsilon_k L_k \leq L_{k-1}/2$, and therefore

$$\frac{5}{4}L_k \leq L_k + \frac{1}{2}L_{k-1} \leq L_{k+1} \leq L_k + L_{k-1} \leq 2L_k. \quad (4.5)$$

The base case is $L_0 = L_1$. Lemma 4.2, applied successively, shows that every prime up to L_k lies in C_{N+k} . The two initial cases follow from the choice of N and retention.

By (4.5), $L_k \geq L(5/4)^{k-1}$ for $k \geq 1$, so $\sum_{k \geq 1} \epsilon_k < \infty$. Set $u_k = L_k/\varphi^k$. Since $\varphi^{-1} + \varphi^{-2} = 1$, (4.4) implies

$$u_{k+1} \geq (1 - \epsilon_k) \left(\frac{u_k}{\varphi} + \frac{u_{k-1}}{\varphi^2} \right).$$

For $m_k = \min(u_{k-1}, u_k)$ this gives $m_{k+1} \geq (1 - \epsilon_k)m_k$. Thus

$$m_k \geq m_1 \prod_{j=1}^{k-1} (1 - \epsilon_j) \geq m_1 \exp \left(-2 \sum_{j \geq 1} \epsilon_j \right) > 0,$$

where $\log(1 - z) \geq -2z$ for $0 \leq z \leq 1/2$. Hence $L_k \gg \varphi^k$. The largest prime at most L_k is $\sim L_k$ by Lemma 3.1, and is at most Q_{N+k} . Consequently $Q_n \gg \varphi^n$. Together with $Q_n \leq M_n \ll \varphi^n$, this proves the proposition. $\square$

Lemma 4.4 (A contracting recurrence). *Suppose a bounded real sequence (x_n) satisfies $x_n + \rho x_{n-1} \rightarrow t$ for some $0 < \rho < 1$. Then $x_n \rightarrow t/(1 + \rho)$.*

Proof. Put $y_n = x_n - t/(1 + \rho)$. Then $y_n = -\rho y_{n-1} + o(1)$. The finite number $L = \limsup_n |y_n|$ satisfies $L \leq \rho L$, so $L = 0$. $\square$

Proposition 4.5. *There are constants $0 < d \leq c$ such that*

$$M_n \sim c\varphi^n, \quad Q_n \sim d\varphi^n. \quad (4.6)$$

Proof. Define

$$T_n = \varphi^{-n} \left(M_n + \frac{M_{n-1}}{\varphi} \right).$$

Using $\varphi - \varphi^{-1} = 1$ and (4.2), we obtain

$$T_{n+1} - T_n = \varphi^{-(n+1)} (M_{n+1} - M_n - M_{n-1}) \leq 0.$$

Proposition 4.3 bounds T_n away from zero, so $T_n \rightarrow t > 0$. Apply Lemma 4.4 to $x_n = M_n/\varphi^n$ and $\rho = \varphi^{-2}$. It gives $M_n/\varphi^n \rightarrow c = t/(1 + \varphi^{-2}) > 0$.

The sequence Q_n is nondecreasing, and Proposition 4.3 bounds Q_n/Q_{n-1} above by a fixed constant. Apply Lemma 4.2 with $X = Q_n$, $Y = Q_{n-1}$. All primes up to $Z = Q_n + Q_{n-1} - Q_n/\log^2 Q_n$ belong to C_{n+1} . Since $Z \asymp Q_n$, Lemma 3.1 shows that the largest prime at most Z is at least $Z - O(Q_n/\log^2 Q_n)$. Therefore

$$Q_{n+1} \geq Q_n + Q_{n-1} - O(Q_n/\log^2 Q_n). \quad (4.7)$$

Set

$$V_n = \varphi^{-n} \left(Q_n + \frac{Q_{n-1}}{\varphi} \right).$$

It is bounded and bounded away from zero. Since $Q_n \asymp \varphi^n$ and $\log Q_n = n \log \varphi + O(1)$, (4.7) gives, for fixed $D > 0$ and all $n \geq n_0$,

$$V_{n+1} - V_n \geq -\frac{D}{n^2}.$$

The sequence $V_n + D \sum_{j=n_0}^{n-1} j^{-2}$ is bounded and nondecreasing, hence convergent. Since the added sums converge, $V_n \rightarrow v > 0$. Lemma 4.4 now gives $Q_n/\varphi^n \rightarrow d = v/(1 + \varphi^{-2}) > 0$. Finally $Q_n \leq M_n$ implies $d \leq c$. $\square$

5 Approximate completeness of the prime prefix

Proposition 5.1. *The constants in Proposition 4.5 are equal: $d = c$. In particular $Q_n/M_n \rightarrow 1$.*

Proof. Assume $d < c$. Choose a fixed integer $k \geq 0$ such that

$$d < T := c\varphi^{-k} \leq \varphi d. \quad (5.1)$$

For example, $k = \lceil \log(c/d)/\log \varphi \rceil - 1$ has these properties. Let

$$X = \varphi^{n-1}, \quad t_n = M_{n-1-k}, \quad \Delta = T - d, \quad \epsilon = \Delta/16.$$

All these constants except X and t_n are now fixed. For sufficiently large n , the integer t_n is an odd prime in C_{n-1} , by Lemma 4.1 and retention. Moreover,

$$t_n = TX + o(X), \quad Q_{n-1} = dX + o(X), \quad Q_n = \varphi dX + o(X). \quad (5.2)$$

Generating primes beyond the predicted prefix. Define

$$\begin{aligned} I_X &= [(\varphi d - 2\epsilon)X, (\varphi d - \epsilon)X], \\ J_X &= [(2d - T + 3\epsilon)X, (2d - T + 9\epsilon)X], \\ U_X &= [(\varphi^2 d + \epsilon)X, (\varphi^2 d + 2\epsilon)X]. \end{aligned} \quad (5.3)$$

All endpoints are positive. In particular, $T \leq \varphi d < 2d$ makes the lower endpoint of J_X positive, and $\epsilon \leq (\varphi - 1)d/16$ makes that of I_X positive. The upper endpoint of I_X is below Q_n for large n , and the upper endpoint of J_X is

$$(2d - T + 9\epsilon)X = (d - 7\epsilon)X < Q_{n-1}.$$

It follows that every prime in I_X belongs to C_n , and every prime in J_X belongs to C_{n-1} .

For an integer $u \in U_X$, consider the target $N = 2u - t_n$. It is odd, and it lies in $[1, BX]$ for a fixed sufficiently large B : its leading lower coefficient is $2\varphi^2 d + 2\epsilon - T > 0$, and its upper coefficient is fixed. For every real $p \in I_X$, the value $q = N - 2p$ satisfies

$$(2d - T + 4\epsilon)X + o(X) \leq q \leq (2d - T + 8\epsilon)X + o(X). \quad (5.4)$$

To verify these bounds, use the lower endpoint of U_X and the upper endpoint of I_X for the lower bound, and the opposite endpoints for the upper bound, together with $\varphi^2 - \varphi = 1$. The error is exactly $-(t_n - TX)$, independent of u and p . For large n its absolute value is less than $\epsilon X/2$. Thus every such q lies in J_X , and

$$\text{meas}\{p \in I_X : 2u - t_n - 2p \in J_X\} = |I_X| = \epsilon X.$$

Lemma 3.3 applies with these fixed intervals and $\eta = \epsilon$. Since $u \mapsto 2u - t_n$ is injective, all but $O_A(X/\log^A X)$ integers $u \in U_X$ have a representation

$$2p + q = 2u - t_n, \quad p \in \mathbb{P} \cap I_X, \quad q \in \mathbb{P} \cap J_X. \quad (5.5)$$

If such a u is itself prime, both operations in

$$b = \frac{t_n + q}{2} = s(t_n + q) \in C_n, \quad u = p + b = s(p + b) \in C_{n+1} \quad (5.6)$$

are valid. Indeed t_n and q are odd primes in C_{n-1} , so their sum is composite with least prime factor 2; and $p \in C_n$, while the final sum u is prime. Consequently

$$\#((\mathbb{P} \cap U_X) \setminus C_{n+1}) \ll_A X / \log^A X. \quad (5.7)$$

Let $r_n = R_{n+2}$, the first prime absent from C_{n+2} . It is the prime immediately after Q_{n+2} . By Lemma 3.1 and Proposition 4.5,

$$r_n = \varphi^3 dX + o(X). \quad (5.8)$$

For a prime $u \in U_X$, put $a = r_n - u$. The identity $\varphi^3 - \varphi^2 = \varphi$ gives

$$(\varphi d - 2\epsilon)X + o(X) \leq a \leq (\varphi d - \epsilon)X + o(X).$$

The errors here are uniform in u . By (5.2) and the fixed positive margins, $1 \leq a \leq Q_n$ for large n .

Lemma 3.1 supplies

$$\#(\mathbb{P} \cap U_X) \sim \frac{\epsilon X}{\log X}$$

candidate primes. At most $O(X / \log^4 X)$ of them are absent from C_{n+1} , by (5.7) with $A = 4$. Among the remaining candidates, at most $H_{n+1}(Q_n)$ have $r_n - u \notin C_{n+1}$, since distinct u give distinct differences. Lemma 3.2 and $Q_n \asymp X$ give

$$H_{n+1}(Q_n) \ll \frac{Q_n}{\log^4 Q_n} \ll \frac{X}{\log^4 X}.$$

The combined exclusions are $o(X / \log X)$, so at least one candidate has both $u \in C_{n+1}$ and $r_n - u \in C_{n+1}$. For that candidate,

$$s(u + (r_n - u)) = r_n \in C_{n+2},$$

because r_n is prime. This contradicts its definition and proves $d = c$. $\square$

6 Cardinalities, limiting density and proof of Theorem 1.1

Completion of the proof of Theorem 1.1. Propositions 4.5 and 5.1 give $M_n \sim Q_n \sim c\varphi^n$. Since both M_{n-1} and Q_{n-1} are integers, Lemma 3.2 gives

$$\begin{aligned} H_n(M_{n-1}) &\leq M_{n-1} - Q_{n-1} + H_n(Q_{n-1}) \\ &\leq M_{n-1} - Q_{n-1} + O\left(\frac{Q_{n-1}}{\log^4 Q_{n-1}}\right) = o(M_{n-1}). \end{aligned}$$

On the other hand, (4.1) implies

$$M_{n-1} - H_n(M_{n-1}) \leq |C_n| \leq M_{n-1} + \pi(M_n).$$

Because $M_n / M_{n-1} \rightarrow \varphi$ and $\pi(M_n) = O(M_n / \log M_n) = o(M_{n-1})$, these two bounds give $|C_n| \sim M_{n-1} \sim c\varphi^{n-1}$. Taking successive ratios and dividing by M_n proves (1.5). $\square$

Remark 6.1. The assertion $Q_n / M_n \rightarrow 1$ says that the *complete* prime prefix reaches a relative distance $o(1)$ from the maximum. For the integers we prove $H_n(M_{n-1}) = o(M_{n-1})$, allowing a sparse set of holes. In particular, the proof does not require the solid core $K_n = \max\{K : \{1, \dots, K\} \subseteq C_n\}$ to satisfy $K_n \sim M_{n-1}$.

A Proof of the interval-restricted binary lemma

We prove Lemma 3.3, retaining its fixed parameters. Only the two analytic estimates stated next are used without proof. All subsequent interval, local-factor, and exceptional-set calculations are included. Write $e(z) = \exp(2\pi iz)$, and use ϕ_E for Euler's totient, to distinguish it from the golden ratio φ .

A.1 The analytic estimates and representation count

The major-arc estimate we use is [7, Lemma 5 (Major arc estimates)] or [11, Proposition 24]: for an interval $\mathcal{I} \subseteq [1, x]$, a reduced fraction a/q , and real ξ , one has, for every $D > 0$,

$$\sum_{m \in \mathcal{I} \cap \mathbb{Z}} \Lambda(m) e((a/q + \xi/q)m) = \frac{\mu(q)}{\phi_E(q)} \int_{\mathcal{I}} e(\xi v/q) dv + O_D \left((q + x|\xi|) \frac{x}{\log^D x} \right). \quad (\text{A.1})$$

Its implicit constant need not be effective. Here μ is the Möbius function and Λ is the von Mangoldt function.

The minor-arc estimate is the following consequence of [6, Lemma 1]: if $(a, q) = 1$, $1 \leq q \leq y$, and $|\theta - a/q| \leq q^{-2}$, then

$$\left| \sum_{p \leq y} (\log p) e(p\theta) \right| \ll (yq^{-1/2} + y^{4/5} + (yq)^{1/2}) (\log(2y))^4. \quad (\text{A.2})$$

The additional factor $1 + q^2|\theta - a/q|$ in the cited lemma is at most 2 under this hypothesis.

Define

$$S_I(\theta) = \sum_{p \in \mathbb{P} \cap I_X} (\log p) e(p\theta), \quad S_J(\theta) = \sum_{q \in \mathbb{P} \cap J_X} (\log q) e(q\theta).$$

Orthogonality of the functions $e(m\theta)$ for integers m gives

$$R_X(N) := \int_0^1 S_I(2\theta) S_J(\theta) e(-N\theta) d\theta = \sum_{\substack{p \in \mathbb{P} \cap I_X, q \in \mathbb{P} \cap J_X \\ 2p+q=N}} (\log p)(\log q). \quad (\text{A.3})$$

Thus $R_X(N) > 0$ is equivalent to a representation of the required kind. Fix

$$K = A + 20, \quad P = (\log X)^K. \quad (\text{A.4})$$

For reduced fractions a/q with $1 \leq q \leq P$ and $0 \leq a < q$, let

$$\mathfrak{M}(a, q) = \{\theta \in \mathbb{R}/\mathbb{Z} : \|\theta - a/q\| \leq P/X\}, \quad \mathfrak{M} = \bigcup_{q \leq P} \bigcup_{\substack{0 \leq a < q \\ (a, q) = 1}} \mathfrak{M}(a, q).$$

Here $\|\cdot\|$ is circular distance to 0; the fraction 0/1 is included. Distinct fractions are at circular distance at least P^{-2} , so these arcs are disjoint once $2P^3 < X$. Their total measure is $O(P^3/X)$. Set $\mathfrak{m} = (\mathbb{R}/\mathbb{Z}) \setminus \mathfrak{M}$, and denote the two parts of (A.3) by $R_{\mathfrak{M}}(N)$ and $R_{\mathfrak{m}}(N)$.

A.2 The minor arcs

Put $D_0 = \lceil X/P \rceil$. Dirichlet approximation gives, for every $\theta \in \mathfrak{m}$, a reduced fraction a/q with

$$1 \leq q \leq D_0, \quad |\theta - a/q| \leq \frac{1}{qD_0} \leq \frac{P}{qX}.$$

Distances may be interpreted modulo 1 by changing a by a multiple of q . If $q \leq P$, this places θ in $\mathfrak{M}(a, q)$, a contradiction. Thus, for large X ,

$$P < q \leq \frac{2X}{P}, \quad |\theta - a/q| \leq q^{-2}. \quad (\text{A.5})$$

Apply (A.2) to the two initial prime sums defining S_J . Their endpoints are fixed positive multiples of X , so q is below both endpoints for all sufficiently large X . An endpoint convention changes the sum by at most $O(\log X)$. Using (A.5), we obtain

$$\sup_{\theta \in \mathfrak{m}} |S_J(\theta)| \ll XP^{-1/2}(\log X)^4 + X^{4/5}(\log X)^4. \quad (\text{A.6})$$

Parseval's identity and (3.3) give

$$\int_0^1 |S_I(2\theta)|^2 d\theta = \sum_{p \in \mathbb{P} \cap I_X} (\log p)^2 \ll X \log X.$$

The factor 2 does not affect orthogonality, since $2(p - p')$ is a nonzero integer when $p \neq p'$. Bessel's inequality, applied to $\mathbf{1}_{\mathfrak{m}}(\theta)S_I(2\theta)S_J(\theta)$, therefore yields

$$\begin{aligned} \sum_{N \in \mathbb{Z}} |R_{\mathfrak{m}}(N)|^2 &\leq \int_{\mathfrak{m}} |S_I(2\theta)S_J(\theta)|^2 d\theta \\ &\ll X^3 P^{-1} (\log X)^9 + X^{13/5} (\log X)^9 \ll_A \frac{X^3}{\log^{A+2} X}. \end{aligned} \quad (\text{A.7})$$

For the last inequality, the first term has logarithmic exponent $9 - K = -A - 11$, and the second satisfies $X^{-2/5} \log^{A+11} X \rightarrow 0$ after division by $X^3 / \log^{A+2} X$. In particular, for each fixed $\lambda > 0$,

$$\#\{N \in \mathbb{Z} : |R_{\mathfrak{m}}(N)| > \lambda X\} \ll_{A,\lambda} \frac{X}{\log^{A+2} X}. \quad (\text{A.8})$$

A.3 The major arcs and the real solution measure

For real ζ , put

$$V_I(\zeta) = \int_{I_X} e(v\zeta) dv, \quad V_J(\zeta) = \int_{J_X} e(v\zeta) dv.$$

On an arc $\theta = a/q + \zeta$, $|\zeta| \leq P/X$, let $q' = q/(q, 2)$, the denominator after reducing $2a/q$. Formula (A.1) implies, for every fixed $E > 0$,

$$\begin{aligned} S_J(a/q + \zeta) &= \frac{\mu(q)}{\phi_E(q)} V_J(\zeta) + O_{E,K}(X/\log^E X), \\ S_I(2a/q + 2\zeta) &= \frac{\mu(q')}{\phi_E(q')} V_I(2\zeta) + O_{E,K}(X/\log^E X). \end{aligned} \quad (\text{A.9})$$

Here is the required uniformity check. Choose $x = CX$ with fixed $C > \max(\beta, \delta, 1)$. In the first line of (A.9), use $\xi = q\zeta$ in (A.1); in the second, use $\xi = 2q'\zeta$ with denominator q' . In both cases $q + x|\xi| = O(P^2)$, with q' replacing q when appropriate. Taking exponent $D = E + 2K + 2$ in (A.1) gives the stated error. Removing prime powers costs at most $O(\sqrt{X} \log^2 X) = O_E(X/\log^E X)$, as in Lemma 3.1.

Take $E = A + 3K + 10$. The sums S_I, S_J and the integrals V_I, V_J are $O(X)$; for the sums, this follows from (3.3). Since the major arcs have measure $O(P^3/X)$, multiplying (A.9) and integrating causes total error

$$O(XP^3/\log^E X) = O(X/\log^{A+10} X). \quad (\text{A.10})$$

Define the Ramanujan sum and its coefficient by

$$c_q(N) = \sum_{\substack{0 \leq a < q \\ (a, q) = 1}} e(-Na/q), \quad w_q = \frac{\mu(q)\mu(q')}{\phi_E(q)\phi_E(q')}.$$

The main term is

$$\sum_{q \leq P} w_q c_q(N) \int_{-P/X}^{P/X} V_I(2\zeta) V_J(\zeta) e(-N\zeta) d\zeta.$$

For any interval $[a, b]$ and nonzero ζ , $|\int_a^b e(v\zeta) dv| \leq \min(b-a, 1/(\pi|\zeta|))$. Consequently

$$|V_I(2\zeta) V_J(\zeta)| \ll \min(X^2, |\zeta|^{-2}), \quad \int_{|\zeta| > P/X} |V_I(2\zeta) V_J(\zeta)| d\zeta \ll X/P.$$

Using $|c_q(N)| \leq \phi_E(q)$, the error from extending every integral to $\mathbb{R}$ is at most

$$\frac{CX}{P} \sum_{q \leq P} \frac{1}{\phi_E(q')} \ll \frac{X \log^2(2P)}{P}. \quad (\text{A.11})$$

Indeed each integer q' arises from at most two values of q , and $r/\phi_E(r) \leq \tau(r)$, where τ counts divisors. Hence

$$\sum_{q \leq P} \frac{1}{\phi_E(q')} \leq 2 \sum_{r \leq P} \frac{\tau(r)}{r} = 2 \sum_{ab \leq P} \frac{1}{ab} \leq 2(1 + \log P)^2.$$

The inequality $r/\phi_E(r) \leq \tau(r)$ follows prime by prime: for $\ell^a \parallel r$, $\ell/(\ell-1) \leq 2 \leq a+1$.

For clarity, Fourier inversion introduces no unspecified geometric constant here. The continuous function

$$f(y) = \left(\frac{1}{2} \mathbf{1}_{2I_X} * \mathbf{1}_{J_X} \right)(y) = \int_{I_X} \mathbf{1}_{J_X}(y-2v) dv$$

has Fourier transform $V_I(2\zeta) V_J(\zeta)$ with the positive exponential convention used above. That product is integrable by the displayed bound, and $f(N) = \mathcal{J}_X(N)$. Fourier inversion gives

$$\int_{\mathbb{R}} V_I(2\zeta) V_J(\zeta) e(-N\zeta) d\zeta = \mathcal{J}_X(N). \quad (\text{A.12})$$

Write

$$\mathfrak{S}_P(N) = \sum_{q \leq P} w_q c_q(N).$$

Combining (A.10)–(A.12), and using $K = A + 20$, proves the uniform formula

$$R_{\mathfrak{M}}(N) = \mathfrak{S}_P(N) \mathcal{J}_X(N) + O_A(X/\log^{A+2} X). \quad (\text{A.13})$$

For the second error, note explicitly that $\log^2(2P)/P = O_A(\log^{-A-2} X)$.

A.4 An elementary bound for the series tail

We first prove that, for $Y \geq 2$ and $N \geq 1$,

$$\sum_{q>Y} \frac{|c_q(N)|}{\phi_E(q)^2} \ll \frac{\log^3(2Y)}{Y} F(N), \quad F(N) = \sum_{h|N} \left(\frac{h}{\phi_E(h)} \right)^2. \quad (\text{A.14})$$

The identity

$$c_q(N) = \sum_{h|(q,N)} h \mu(q/h) \quad (\text{A.15})$$

follows by inserting $\mathbf{1}_{(a,q)=1} = \sum_{d|(a,q)} \mu(d)$ into the defining exponential sum. The sum over multiples of d is q/d if $q/d \mid N$ and zero otherwise; set $h = q/d$. In particular, $|c_q(N)| \leq \sum_{h|(q,N)} h$.

The product formula for the totient gives $\phi_E(hk) \geq \phi_E(h)\phi_E(k)$. Therefore, writing $q = hk$ and interchanging nonnegative sums,

$$\sum_{q>Y} \frac{|c_q(N)|}{\phi_E(q)^2} \leq \sum_{h|N} \frac{h}{\phi_E(h)^2} \sum_{k>Y/h} \frac{1}{\phi_E(k)^2}. \quad (\text{A.16})$$

For $z \geq 1$ we claim

$$\sum_{k>z} \frac{1}{\phi_E(k)^2} \ll \frac{\log^3(2z)}{z}. \quad (\text{A.17})$$

Let $\tau_4(k)$ count ordered factorizations $k = abcd$ in positive integers. Prime by prime,

$$\left(\frac{k}{\phi_E(k)} \right)^2 \leq \tau(k)^2 \leq \tau_4(k),$$

since $(a+1)^2 \leq \binom{a+3}{3}$ for every integer $a \geq 0$. Furthermore, for $w \geq 1$,

$$\sum_{k \leq w} \tau_4(k) = \sum_{abc \leq w} \lfloor \frac{w}{abc} \rfloor \leq w \left(\sum_{a \leq w} \frac{1}{a} \right)^3 \leq w(1 + \log w)^3.$$

Split $k > z$ into ranges $2^j z < k \leq 2^{j+1} z$. These inequalities give

$$\sum_{k>z} \frac{1}{\phi_E(k)^2} \leq \sum_{j \geq 0} \frac{1}{(2^j z)^2} \sum_{k \leq 2^{j+1} z} \tau_4(k) \ll \frac{1}{z} \sum_{j \geq 0} 2^{-j} (\log(2z) + j)^3 \ll \frac{\log^3(2z)}{z},$$

because $\sum_{j \geq 0} 2^{-j} (j+1)^3$ converges. This proves (A.17). For $0 \leq z < 1$, the same sum is $O(1)$, by the bound at $z = 1$ and the single term $k = 1$.

For $h \leq Y$, apply (A.17) to $z = Y/h$ in (A.16), bounding $\log(2Y/h)$ by $\log(2Y)$. The resulting contribution for h is at most

$$C \frac{\log^3(2Y)}{Y} \frac{h^2}{\phi_E(h)^2}.$$

For $h > Y$, the $O(1)$ bound gives at most $Ch/\phi_E(h)^2$, which is also bounded by the same expression after increasing C . This proves (A.14), including absolute convergence for each fixed N .

We will also need its first moment. For large X ,

$$\begin{aligned} \sum_{1 \leq N \leq BX} F(N) &\leq BX \sum_{h \leq BX} \frac{1}{h} \left(\frac{h}{\phi_E(h)} \right)^2 \\ &\leq BX \sum_{h \leq BX} \frac{\tau_4(h)}{h} = BX \sum_{abcd \leq BX} \frac{1}{abcd} \leq BX(1 + \log(BX))^4 \ll_B X \log^4 X. \end{aligned} \quad (\text{A.18})$$

A.5 The local factors and positivity

For odd N , absolute convergence from (A.14) allows us to compute the full series $\mathfrak{S}(N) = \sum_{q \geq 1} w_q c_q(N)$. A nonzero coefficient requires q squarefree. For odd squarefree q , we have $q' = q$ and $w_q = \mu(q)^2 / \phi_E(q)^2$. For even squarefree $q = 2r$, with r odd, we have $q' = r$ and

$$w_{2r} = -\frac{\mu(r)^2}{\phi_E(r)^2}, \quad c_{2r}(N) = c_2(N)c_r(N) = -c_r(N).$$

Here $c_q(N)$ is multiplicative in q by (A.15), and $c_2(N) = -1$ for odd N . Thus the even term equals the corresponding odd term, and

$$\begin{aligned} \mathfrak{S}(N) &= 2 \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{\mu(r)^2 c_r(N)}{\phi_E(r)^2} \\ &= 2 \prod_{\ell > 2} \left(1 - \frac{1}{(\ell-1)^2}\right) \prod_{\substack{\ell | N \\ \ell > 2}} \frac{\ell-1}{\ell-2}. \end{aligned} \tag{A.19}$$

All products indexed by ℓ here are over primes. To verify each factor, (A.15) gives $c_\ell(N) = \ell-1$ if $\ell \mid N$ and $c_\ell(N) = -1$ otherwise. The factor at a divisor prime is $1 + 1/(\ell-1)$; dividing this by $1 - 1/(\ell-1)^2$ gives $(\ell-1)/(\ell-2)$, as displayed.

In particular,

$$\mathfrak{S}(N) \geq s_0 := 2 \prod_{\ell > 2} \left(1 - \frac{1}{(\ell-1)^2}\right) > 0 \quad (N \text{ odd}). \tag{A.20}$$

For positivity of this constant, the factors lie in $(0, 1)$, $1/(\ell-1)^2 \leq 1/4$, and $\sum_{\ell > 2} 1/(\ell-1)^2 < \infty$. The inequality $\log(1-z) \geq -2z$ for $0 \leq z \leq 1/2$ bounds the product away from zero.

The odd terms of $\mathfrak{S}_P(N)$ have $r \leq P$, whereas its even terms have $r \leq P/2$. Consequently, for odd N ,

$$|\mathfrak{S}(N) - \mathfrak{S}_P(N)| \leq 2 \sum_{r > P/2} \frac{|c_r(N)|}{\phi_E(r)^2} \ll \frac{\log^3(2P)}{P} F(N).$$

Summing over $1 \leq N \leq BX$ and using (A.18), we see that the number of odd N in this range for which $|\mathfrak{S}(N) - \mathfrak{S}_P(N)| > s_0/2$ is at most

$$O_B \left(\frac{X \log^4 X \log^3(2P)}{P} \right) = O_A \left(\frac{X}{\log^A X} \right). \tag{A.21}$$

The last estimate follows from $K = A + 20$, since $\log^3(2P) = O_A((\log \log X)^3)$.

A.6 Completion of the representation lemma

Outside the exceptional set in (A.21), every odd target has $\mathfrak{S}_P(N) \geq s_0/2$. If also $\mathcal{J}_X(N) \geq \eta X$, then (A.13) gives, for sufficiently large X ,

$$\operatorname{Re} R_{\mathfrak{M}}(N) \geq \frac{s_0 \eta X}{2} - \frac{s_0 \eta X}{8} = \frac{3s_0 \eta X}{8}.$$

By (A.8) with $\lambda = s_0 \eta / 8$, all but $O_A(X / \log^{A+2} X)$ further targets satisfy $|R_{\mathfrak{m}}(N)| \leq s_0 \eta X / 8$. For a target outside both exceptional sets, (A.3) is a nonnegative real number and obeys

$$R_X(N) = \operatorname{Re}(R_{\mathfrak{M}}(N) + R_{\mathfrak{m}}(N)) \geq \frac{s_0 \eta X}{4} > 0.$$

The union of the exceptional sets has size $O_A(X/\log^A X)$. This proves Lemma 3.3. All interval data were fixed before X tended to infinity; no uniformity as an interval length or margin tends to zero has been used. $\square$

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used GPT-6 Astra in order to assist with the construction of the mathematical argument and the drafting of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. The main theorem has been independently checked in Lean 4, as recorded in [4].

Declarations of interest

Declarations of interest: none.

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